

Exercise Sheet 8

Discrete Mathematics by Hongfei Fu, 2022.10.18

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1. ([R], Page 617, Exercise 54) Suppose that R_1 and R_2 are equivalence relations on a set A . Let P_1 and P_2 be the partitions that correspond to R_1 and R_2 , respectively. Show that $R_1 \subseteq R_2$ if and only if P_1 is a refinement of P_2 .

Refinement: A partition P_1 is called a refinement of the partition P_2 if every set in P_1 is a subset of one of the sets in P_2 .

Note: In Exercise 54, the correspondence between equivalence relations and partitions are from Theorem 2 on Page 613 of [R].

Answer Area:

- Suppose that $R_1 \subseteq R_2$.
 - $\therefore$ For an arbitrary $B \in P_1, \forall a, b \in B, aR_1b$.
 - $\therefore R_1 \subseteq R_2$.
 - $\therefore aR_2b$.
 - $\therefore \exists B' \in P_2, a, b \in B'$.
 - $\therefore P_1$ is a partition on A .
 - $\therefore \forall c \in B (c \neq a, c \neq b), cR_1a$.
 - $\therefore cR_2a. \exists B'' \in P_2, a, c \in B''$
 - $\therefore P_2$ is a partition.
 - $\therefore B' = B''$.
 - $\therefore \forall x \in B, \exists B' \in P_2, x \in B'$. Then $B \subseteq B'$
 - $\therefore \forall B \in P_1, \exists B' \in P_2, B \subseteq B'$.
 - $\therefore P_1$ is a refinement of P_2 .
- Suppose that P_1 is a refinement of P_2 .
 - $\therefore \forall a, b (aR_1b), \exists B \in P_1, a, b \in B$.
 - $\therefore P_1$ is a refinement of P_2 .
 - $\therefore \exists B' \in P_2, B \subseteq B'$.
 - $\therefore a, b \in B'$.
 - $\therefore aR_2b$
 - $\therefore R_1 \subseteq R_2$.

2. ([R], Page 617, Exercise 56(a)(b)(c)) Suppose that R_1 and R_2 are equivalence relations on the set S . Determine whether each of these combinations of R_1 and R_2 must be an equivalence relation.

a) $R_1 \cup R_2$ b) $R_1 \cap R_2$ c) $(R_1 \circ R_2) \cup (R_2 \circ R_1)$ (2pt)

Note: In Exercise 56, you need to either prove the result rigorously or give a counterexample.

Answer Area:

a) NO. For example:

R_1, R_2 are relations on $S = \{1, 2, 3, 4\}$.

$R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}$.

$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4), (2, 3), (3, 2)\}$.

$R_1 \cup R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (2, 3), (3, 2)\}$.

Then $(1, 2) \in R_1 \cup R_2$ and $(2, 3) \in R_1 \cup R_2$, but $(1, 3) \notin R_1 \cup R_2$.

So $R_1 \cup R_2$ is not transitive. Then $R_1 \cup R_2$ is not an equivalence relation.

b) Yes. A proof is as below:

$\therefore R_1$ and R_2 are equivalence relations on the set S .
 $\therefore \forall a \in S, aR_1a$ and aR_2a .
 $\therefore \forall a \in S, (a, a) \in R_1 \cap R_2$.
 $\therefore R_1 \cap R_2$ is reflexive.
 $\therefore \forall a \forall b \in S((a, b) \in R_1 \cap R_2), aR_1b$ and aR_2b .
 $\therefore R_1$ and R_2 are symmetric.
 $\therefore bR_1a$ and bR_2a .
 $\therefore (b, a) \in R_1 \cap R_2$.
 $\therefore R_1 \cap R_2$ is symmetric.
 $\therefore \forall a, b, c \in S(((a, b) \in R_1 \cap R_2) \cap ((b, a) \in R_1 \cap R_2)), aR_1b \cap bR_1c$ and $aR_2b \cap bR_2c$.
 $\therefore R_1, R_2$ are transitive.
 $\therefore aR_1c$ and aR_2c .
 $\therefore (a, c) \in R_1 \cap R_2$.
 $\therefore R_1 \cap R_2$ is transitive.
 $\therefore R_1 \cap R_2$ is an equivalence relation.

c) No. For example:

R_1, R_2 are relations on $S = \{1, 2, 3, 4, 5\}$.

$R_1 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (4, 5), (5, 4)\}$.

$R_2 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (1, 2), (2, 1), (4, 3), (3, 4)\}$.

$(R_1 \circ R_2) = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (4, 5), (5, 4), (1, 3), (4, 2), (3, 5)\}$.

$(R_2 \circ R_1) = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (1, 2), (2, 1), (4, 3), (3, 4), (2, 4), (3, 1), (5, 3)\}$.

$(R_1 \circ R_2) \cup (R_2 \circ R_1) = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (2, 3), (3, 2), (4, 5), (5, 4), (1, 3), (4, 2), (3, 5), (1, 2), (2, 1), (4, 3), (3, 4), (2, 4), (3, 1), (5, 3)\}$.

Then $(1, 3) \in (R_1 \circ R_2) \cup (R_2 \circ R_1)$ and $(3, 5) \in (R_1 \circ R_2) \cup (R_2 \circ R_1)$, but $(1, 5) \notin (R_1 \circ R_2) \cup (R_2 \circ R_1)$.

So, $(R_1 \circ R_2) \cup (R_2 \circ R_1)$ is not transitive. Then $(R_1 \circ R_2) \cup (R_2 \circ R_1)$ is not an equivalence relation.

3. ([R], Page 617, Exercise 57) Consider the equivalence relation from Example 2, namely, $R = \{(x, y) | x - y \text{ is an integer}\}$

a) What is the equivalence class of 1 for this equivalence relation?

b) What is the equivalence class of $1/2$ for this equivalence relation?

Answer Area:

a) $[1]_R = \mathbf{Z}$.

b) $[\frac{1}{2}]_R = \{\frac{1}{2} + k | k \in \mathbf{Z}\}$.

4. ([R], Page 631, Exercise 37 (2pt)) Show that lexicographic order is a partial ordering on the Cartesian product of two posets.

Answer Area:

Suppose that we have two posets $(A_1, \preceq_1), (A_2, \preceq_2)$. And we have the lexicographic order $\preceq$ on $A_1 \times A_2$.

According to the definition of the $\preceq$:

$\therefore \forall (a_1, b_1) \in A_1 \times A_2, a_1 = a_1, b_1 = b_2$.

$\therefore (a_1, b_1) \preceq (a_1, b_1)$.

$\therefore \preceq$ is reflexive.

$\therefore \forall a_1, a_2, b_1, b_2 \in A_1 \cup A_2$, if $((a_1, a_2) \preceq (b_1, b_2)) \cap ((b_1, b_2) \preceq (a_1, a_2))$, then:

- $(a_1, a_2) = (b_1, b_2)$.
 - $(a_1, a_2) \prec (b_1, b_2)$ and $(b_1, b_2) \prec (a_1, a_2)$. Then there are two possibilities:
 - $a_1 \prec_1 b_1 \cap b_1 \prec_1 a_1$.
 $\therefore \prec_1$ is a partial order, so it's antisymmetric.
 $\therefore a_1 = b_1$. Then $a_2 \prec_2 b_2 \cap b_2 \prec_2 a_2$.
 $\therefore \prec_2$ is a partial order, so it's antisymmetric.
 $\therefore a_2 = b_2$.
 Then $(a_1, a_2) = (b_1, b_2)$.
 - $a_1 = b_1$ and $a_2 \prec_2 b_2$. The same as above.
- $\therefore (a_1, a_2) = (b_1, b_2)$.

$\therefore \preceq$ is antisymmetric.

$\therefore \forall a_1, a_2, b_1, b_2, c_1, c_2 \in A_1 \times A_2$, if $(a_1, a_2) \preceq (b_1, b_2)$ and $(b_1, b_2) \preceq (c_1, c_2)$. Then:

- $(a_1, a_2) = (b_1, b_2)$ Then $(a_1, a_2) \preceq (c_1, c_2)$.
- $(b_1, b_2) = (c_1, c_2)$ Then $(a_1, a_2) \preceq (c_1, c_2)$.
- $(a_1, a_2) \prec (b_1, b_2)$ and $(b_1, b_2) \prec (c_1, c_2)$. Then:
 - $a_1 \prec_1 b_1$ and $b_1 \prec_1 c_1$.
 $\therefore \prec_1$ is a partial order, so it's transitive.
 $\therefore a_1 \prec_1 c_1$. Then $(a_1, a_2) \prec (c_1, c_2)$. Then $(a_1, a_2) \preceq (c_1, c_2)$.
 - $a_1 = b_1$ or $b_1 = c_1$ (Not both.)
 $\therefore a_1 \prec_1 c_1$. Then $(a_1, a_2) \prec (c_1, c_2)$. Then $(a_1, a_2) \preceq (c_1, c_2)$.
 - $a_1 = b_1 = c_1$
 $\therefore a_2 \prec_2 b_2$ and $b_2 \prec_2 c_2$.
 $\therefore \prec_2$ is a partial order, so it's transitive.
 $\therefore a_2 \prec_2 c_2$.
 $\therefore (a_1, a_2) \prec (c_1, c_2)$. Then $(a_1, a_2) \preceq (c_1, c_2)$.

$\therefore \preceq$ is transitive.

$\therefore$ lexicographic order is a partial ordering on the Cartesian product of two posets.

5. ([R], Page 631, Exercise 39 (**2pt**)) Suppose that $(S, \preceq_1)$ and $(T, \preceq_2)$ are posets. Show that $(S \times T, \preceq)$ is a poset where $(s, t) \preceq (u, v)$ if and only if $s \preceq_1 u$ and $t \preceq_2 v$

Answer Area:

- $\therefore (S, \preceq_1)$ and $(T, \preceq_2)$ are posets.
- $\therefore \preceq_1, \preceq_2$ are partial orders. Then they are reflexive.
- $\therefore s \preceq_1 s$ and $t \preceq_2 t$.
- $\therefore (s, t) \preceq (s, t)$.
- $\therefore \preceq$ is reflexive.
- $\therefore$ If $(s, t) \preceq (u, v)$ and $(u, v) \preceq (s, t)$, then $s \preceq_1 u \cap u \preceq_1 s$ and $t \preceq_2 v \cap v \preceq_2 s$.
- $\therefore \preceq_1$ and $\preceq_2$ are antisymmetric.
- $\therefore s = u$ and $t = v$.
- $\therefore (s, v) = (t, u)$
- $\therefore \preceq$ is antisymmetric.

$\therefore$ If $(s, v) \preceq (u, t)$ and $(u, t) \preceq (x, y)$, then $s \preceq_1 u \cap u \preceq_1 x$ and $v \preceq_2 t \cap t \preceq_2 y$.
 $\therefore \preceq_1$ and $\preceq_2$ are transitive.
 $\therefore s \preceq_1 x$ and $v \preceq_2 y$.
 $\therefore (s, v) \preceq (x, y)$.
 $\therefore \preceq$ is transitive.
 $\therefore \preceq$ is a partial order.
 $\therefore (S \times T, \preceq)$ is a poset.

6. ([R], Page 632, Exercise 57) Show that the poset of rational numbers with the usual "less than or equal to" relation, $(\mathbf{Q}, \leq)$, is a dense poset.

Definition of dense: A poset $(R, \preceq)$ is **dense** if for all $x \in R$ and $y \in R$ with $x \prec y$, there is an element $z \in R$ such that $x \prec z \prec y$.

Answer Area:

$\therefore \prec := a \preceq b \cap a \neq b$. $\therefore$ If $\preceq := \leq$, then $\prec := a \leq b \cap a \neq b$. That is $\prec := \leq$
 $\forall x \in \mathbf{Q}, y \in \mathbf{Q}, x \prec y$ (i.e. $x < y$), $\exists z = \frac{x+y}{2} \in \mathbf{Q}, x < z < y$. Then $x \prec z \prec y$.
 So, $(\mathbf{Q}, \leq)$ is a dense poset.